2 Probability

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2.5 Independence

Independence

The definition of conditional probability enables us to revise the probability $P(A)$ originally assigned to A when we are subsequently informed that another event B has occurred; the new probability of A is $P(A \mid B)$.

In our examples, it was frequently the case that $P(A \mid B)$ differed from the unconditional probability $P(A)$, indicating that the information "B has occurred" resulted in a change in the chance of A occurring.

Often the chance that A will occur or has occurred is not affected by knowledge that B has occurred, so that $P(A \mid B) = P(A)$.

Independence

It is then natural to regard A and B as independent events, meaning that the occurrence or nonoccurrence of one event has no bearing on the chance that the other will occur.

Definition

Two events A and B are **independent** if $P(A\,|\,B) = P(A)$ and are **dependent** otherwise.

The definition of independence might seem "unsymmetric" because we do not also demand that $P(B\,|\,A) = P(B)$.

Independence

However, using the definition of conditional probability and the multiplication rule,

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \mid B)P(B)}{P(A)} \qquad \textbf{(2.7)}$$

The right-hand side of Equation (2.7) is $P(B)$ if and only if $P(A \mid B) = P(A)$ (independence), so the equality in the definition implies the other equality (and vice versa).

It is also straightforward to show that if A and B are independent, then so are the following pairs of events: (1) A' and B, (2) A and B', and (3) A' and B'.

Example 32

Consider a gas station with six pumps numbered 1, 2, . . . , 6, and let E_i denote the simple event that a randomly selected customer uses pump i (i = 1, . . . ,6).

Suppose that

$$P(E_1) = P(E_6) = .10,$$
$$P(E_2) = P(E_5) = .15,$$
$$P(E_3) = P(E_4) = .25$$

Define events A, B, C by

$$A = \{2, 4, 6\},\ B = \{1, 2, 3\},\ C = \{2, 3, 4, 5\}.$$

Example 32

cont'd

We then have $P(A) = .50$, $P(A \mid B) = .30$, and $P(A \mid C) = .50$. That is, events A and B are dependent, whereas events A and C are independent.

Intuitively, A and C are independent because the relative division of probability among even- and odd-numbered pumps is the same among pumps 2, 3, 4, 5 as it is among all six pumps.

The Multiplication Rule for $P(A \cap B)$

The Multiplication Rule for $P(A \cap B)$

Frequently the nature of an experiment suggests that two events A and B should be assumed independent.

This is the case, for example, if a manufacturer receives a circuit board from each of two different suppliers, each board is tested on arrival, and

A = {first is defective} and

B = {second is defective}.

The Multiplication Rule for $P(A \cap B)$

If $P(A) = .1$, it should also be the case that $P(A \mid B) = .1$; knowing the condition of the second board shouldn't provide information about the condition of the first.

The probability that both events will occur is easily calculated from the individual event probabilities when the events are independent.

The Multiplication Rule for $P(A \cap B)$

Proposition

A and *B* are independent if and only if (iff)

$$P(A \cap B) = P(A) \cdot P(B) \tag{2.8}$$

The verification of this multiplication rule is as follows:

$$P(A \cap B) = P(A \mid B) \cdot P(B) = P(A) \cdot P(B) \tag{2.9}$$

where the second equality in Equation (2.9) is valid iff *A* and *B* are independent. Equivalence of independence and Equation (2.8) imply that the latter can be used as a definition of independence.

Example 34

It is known that 30% of a certain company's washing machines require service while under warranty, whereas only 10% of its dryers need such service.

If someone purchases both a washer and a dryer made by this company, what is the probability that both machines will need warranty service?

Example 34

cont'd

Let A denote the event that the washer needs service while under warranty, and let B be defined analogously for the dryer.

Then $P(A) = .30$ and $P(B) = .10$.

Assuming that the two machines will function independently of one another, the desired probability is

$$P(A \cap B) = P(A) \cdot P(B) = (.30)(.10) = .03$$

Independence of More Than Two Events

Independence of More Than Two Events

The notion of independence of two events can be extended to collections of more than two events.

Although it is possible to extend the definition for two independent events by working in terms of conditional and unconditional probabilities, it is more direct and less cumbersome to proceed along the lines of the last proposition.

Independence of More Than Two Events

Definition

Events $A_1, \dots, A_n$ are **mutually independent** if for every k ($k = 2, 3, \dots, n$) and every subset of indices $i_1, i_2, \dots, i_k$,

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \cdot P(A_{i_2}) \cdot \cdots \cdot P(A_{i_k})$$

Events $A_1, \dots, A_n$ are **mutually independent** if for every k ($k = 2, 3, \dots, n$) and every subset of indices $i_1, i_2, \dots, i_k$,

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \cdot P(A_{i_2}) \cdot \cdots \cdot P(A_{i_k})$$

Independence of More Than Two Events

To paraphrase the definition, the events are mutually independent if the probability of the intersection of any subset of the *n* events is equal to the product of the individual probabilities.

In using the multiplication property for more than two independent events, it is legitimate to replace one or more of the A_is by their complements (e.g., if A_1, A_2, and A_3 are independent events, so are A_1', A_2', and A_3').

Independence of More Than Two Events

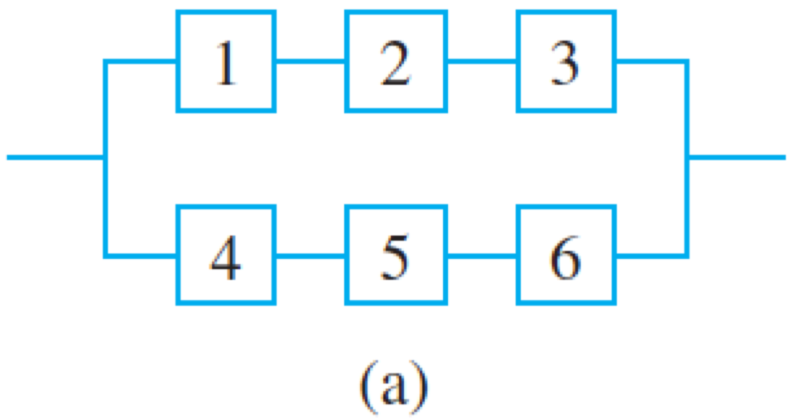
As was the case with two events, we frequently specify at the outset of a problem the independence of certain events.

The probability of an intersection can then be calculated via multiplication.

Example 36

The article “Reliability Evaluation of Solar Photovoltaic Arrays”(*Solar Energy*, 2002: 129–141) presents various configurations of solar photovoltaic arrays consisting of crystalline silicon solar cells.

Consider first the system illustrated in Figure 2.14(a).



System configurations for Example 36: (a) series-parallel

Figure 2.14(a)

Example 36

cont'd

There are two subsystems connected in parallel, each one containing three cells.

In order for the system to function, at least one of the two parallel subsystems must work.

Within each subsystem, the three cells are connected in series, so a subsystem will work only if all cells in the subsystem work.

Example 36

cont'd

Consider a particular lifetime value t_0, and supose we want to determine the probability that the system lifetime exceeds t_0.

Let A_i denote the event that the lifetime of cell *i* exceeds t_0 (i = 1, 2, . . . , 6).

We assume that the $A_i's$ are independent events (whether any particular cell lasts more than t_0 hours has no bearing on whether or not any other cell does) and that $P(A_i) = .9$ for every *i* since the cells are identical.

Example 36

cont'd

Then

P(system lifetime exceeds t_0)

$$= P[(A_1 \cap A_2 \cap A_3) \cup (A_4 \cap A_5 \cap A_6)]$$

$$= P(A_1 \cap A_2 \cap A_3) + P(A_4 \cap A_5 \cap A_6) - P[(A_1 \cap A_2 \cap A_3) \cap (A_4 \cap A_5 \cap A_6)]$$

$$= (.9)(.9)(.9) + (.9)(.9)(.9) - (.9)(.9)(.9)(.9)(.9)(.9)$$

$$= .927$$

Example 36

cont'd

Alternatively,

P(system lifetime exceeds t_0)

$= 1 - P(\text{both subsystem lives are} \le t_0)$

$= 1 - [P(\text{subsystem life is} \le t_0)]^2$

$= 1 - [1 - P(\text{subsystem life is} > t_0)]^2$

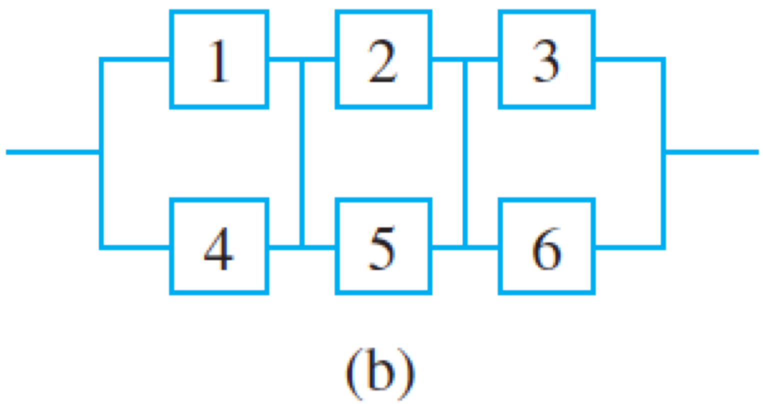
$= 1 - [1 - (.9)^3]^2$

$= .927$

Example 36

cont'd

Next consider the total-cross-tied system shown in Figure 2.14(b), obtained from the series-parallel array by connecting ties across each column of junctions. Now the system fails as soon as an entire column fails, and system lifetime exceeds t_0 only if the life of every column does so. For this configuration,



System configurations for Example 36: (b) total-cross-tied

Figure 2.14(b)

Example 36

cont'd

P(system lifetime is at least t_0)

$= [P(\text{column lifetime exceeds } t_0)]^3$

$= [1 - P(\text{column lifetime} \le t_0)]^3$

$= [1 - P(\text{both cells in a column have lifetime} \le t_0)]^3$

$= [1 - (1 - .9)^2]^3$

$= .970$